SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

Dr.G.Ayyappan

QUESTIONS		Total Marks: 10	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	**1**
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	**1**
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	**1**
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	**1**
5	Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	**1**
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	**1**
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	**1**

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**1**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**1**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**1**

Code of Conduct:

I V. Sujithra (192521445) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

MCQ Answers

1. Which of the following is true about 1NF?
Answer: **(c) All attributes must be atomic**

2. A relation is in 2NF if it is in 1NF and:
Answer: **(b) Has no partial dependencies on primary key**

3. Which normal form deals with multi-valued dependencies?
Answer: **(d) 4NF**

4. BCNF is a stricter version of:
Answer: **(c) 3NF**

5. Functional dependency $X \rightarrow Y$ means:
Answer: **(b) X determines Y**

6. Which of the following anomaly is NOT associated with unnormalized relations?
Answer: **(d) Retrieval anomaly**

7. A superkey is:
Answer: **(b) Any set of attributes that uniquely identifies tuples**

8. If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:
Answer: **(c) Transitively dependent on A**

9. 5NF is also known as:
Answer: **(c) Project-Join Normal Form**

10. Lossless-join decomposition ensures:
Answer: **(a) No data is lost during decomposition**

Final Answer Key

Q.NO	Correct Answer	Explanation
1	**(c) All attributes must be atomic**	In First Normal Form (1NF), every attribute must contain only a single (atomic) value. Multi-valued or repeating attributes are not allowed.
2	**(b) Has no partial dependencies on primary key**	A relation is in 2NF if it is in 1NF and every non-key attribute is fully dependent on the entire primary key.
3	**(d) 4NF**	Fourth Normal Form (4NF) removes multi-valued dependencies from a relation.
4	**(c) 3NF**	BCNF is a stricter version of Third Normal Form (3NF) because every determinant must be a super key.
5	**(b) X determines Y**	A functional dependency $X \rightarrow Y$ means that the value of X uniquely determines the value of Y.
6	**(d) Retrieval anomaly**	Insertion, Update, and Deletion anomalies are common in unnormalized relations. Retrieval anomaly is not considered a database normalization anomaly.
7	**(b) Any set of attributes that uniquely identifies tuples**	A Super Key is any combination of attributes that uniquely identifies each tuple in a relation.
8	**(c) Transitively dependent on A**	Since $A \rightarrow B$ and $B \rightarrow C$, attribute C is transitively dependent on A.
9	**(c) Project-Join Normal Form**	Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJNF).
10	**(a) No data is lost during decomposition**	A Lossless-Join Decomposition ensures that the original relation can be reconstructed without losing any information.